Linear Regression

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Outline

The Linear Model
- Simple linear regression
- History
- Multiple linear regression

Optimisation algorithms
- Gradient Descent
- Least-Squares

Regression libraries in Python
- sklearn
- statsmodels

Problems
- Interpretation Problem parameters
- Exercises

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Simple linear regression

Input and output

- Data pairs (x^t, y^t), $t = 1, \ldots, T$.
- Input $x^t \in \mathbb{R}$
- Output $y^t \in \mathbb{R}$.

Modelling the conditional expectation $\mathbb{E}[y^t|x^t]$

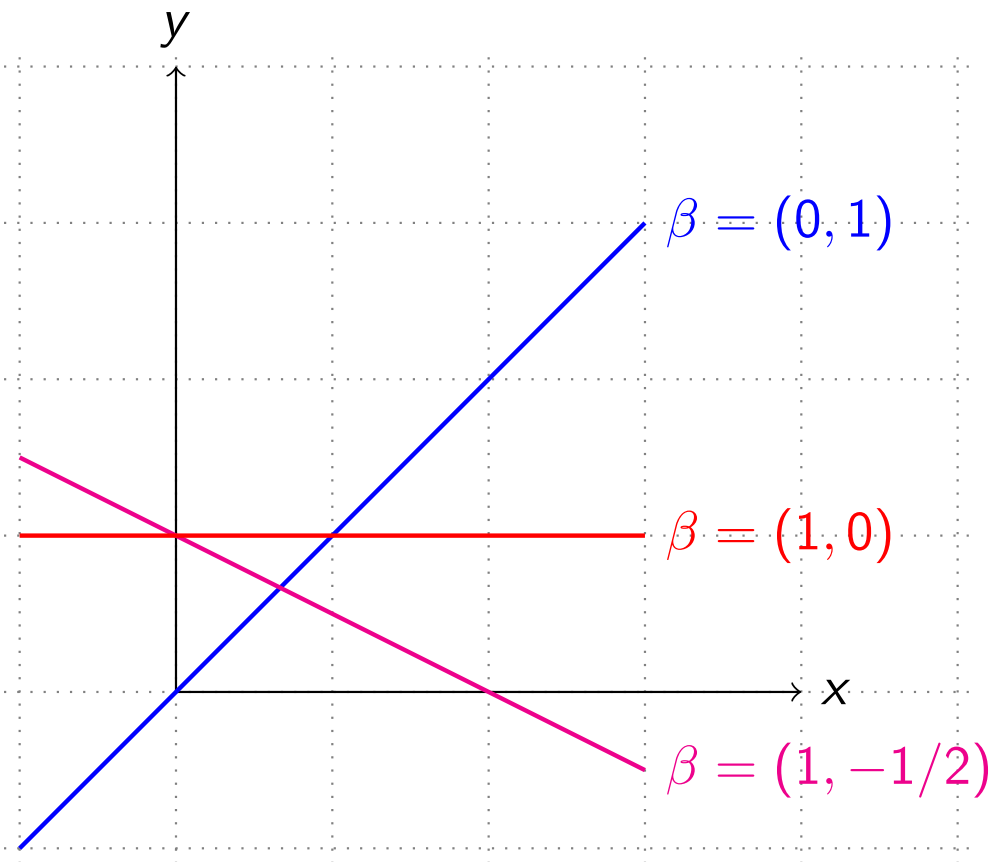
- Parameters $\beta_0, \beta_1 \in \mathbb{R}$
- Function $\pi_\beta : \mathbb{R} \to \mathbb{R}$, defined as

$$\pi_\beta(x^t) = \beta_0 + \beta_1 x_t$$

Probabilistic predictions: Modelling the conditional probability $\mathbb{P}[y^t|x^t]$

- $y^t = \mathbb{E}[y^t|x^t] + \epsilon_t$, with $\epsilon_t \in \mathbb{R}$ being zero-mean noise.
- Simplest model: variance $\sigma = \mathbb{V}(\epsilon) = \mathbb{E}[\epsilon_t^2]$

Linear models



$$\pi_\beta(x) = \beta_0 + \beta_1 x = [\beta_0, \beta_1] \begin{bmatrix} 1 \\ x \end{bmatrix}$$

Two views of the problem

Learning as optimisation

- Each value $\pi_\beta(x)$ is a prediction about the value of y

Learning as inference

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- Each value $\pi_\beta(x)$ is a prediction about the value of y
- We suffer a loss $\ell(y, \pi_\beta(x))$ for every example (x, y) that we see.

Learning as inference

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- We can minimise the average loss over the data

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Learning as inference

- The parameters β define a probabilistic model $P_\beta(y|x)$ for every value of y.

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- We want to find the parameters giving the highest probability on the observed data.

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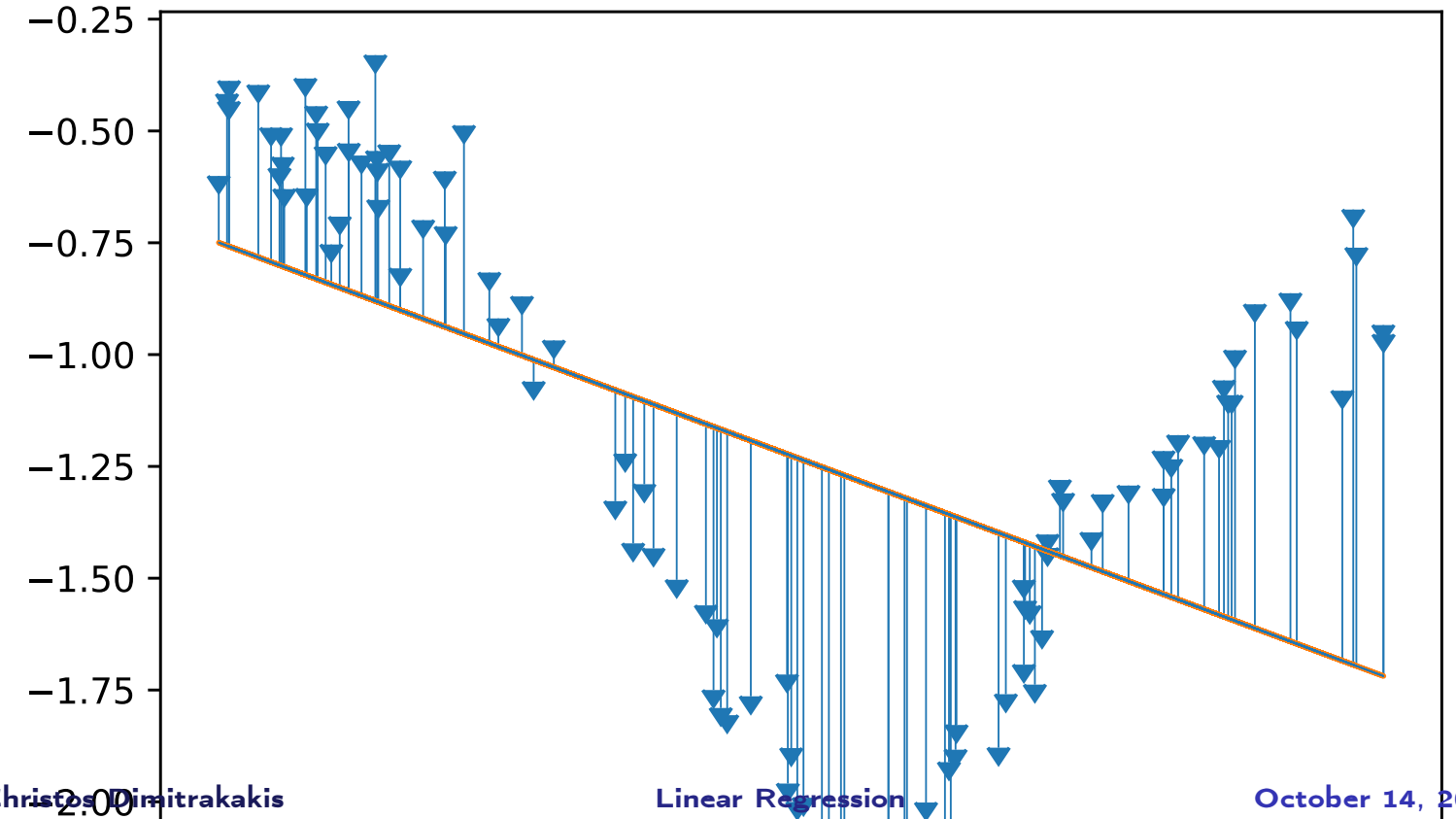
Learning as inference

- The parameters β define a probabilistic model $P_\beta(y|x)$ for every value of y.
- We want to find the parameters giving the highest probability on the observed data.
- Ideally, we want to find the true conditional distribution $P(y|x)$.

Learning as Optimisation

Find the parameters β minimising squared error

$$\min_{\beta} \frac{1}{T} \sum_{t=1}^{T} \Big[\underbrace{y^t - \pi_\beta(x^t)}_{\text{residual}} \Big]^2$$



Origins

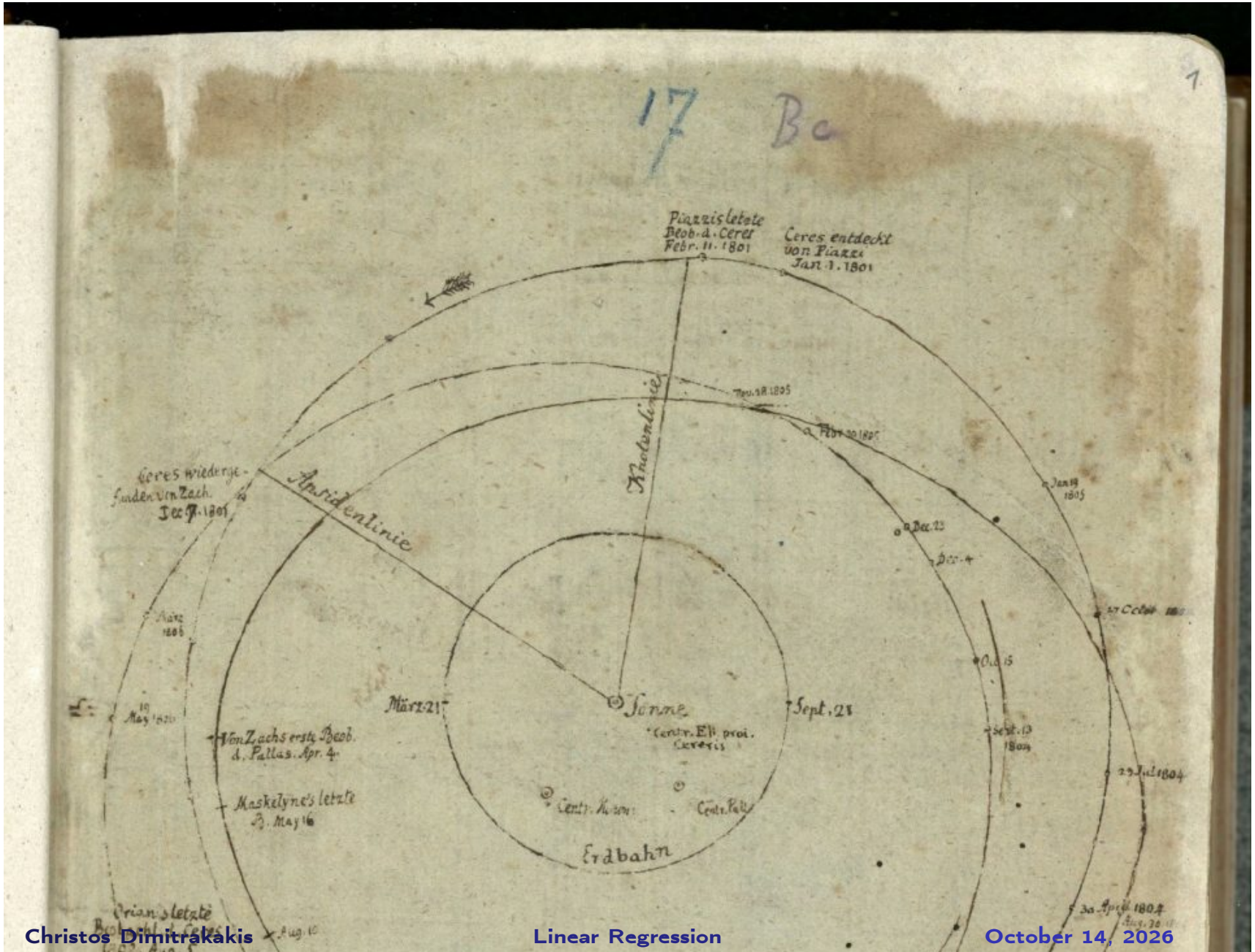


Figure: Gauss: originator

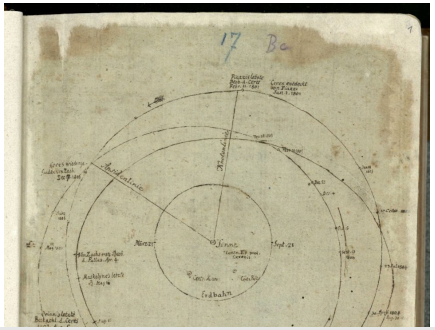


Figure: Legendre: first publication

The orbit of Ceres



The orbit of Ceres



On Gauss and Legendre. Gauss applied the method of least squares to calculate the orbit of Ceres, an asteroid detected in the solar system. At the time, astronomers were keen to establish what the orbit was, but they did not even know where to point their telescopes. He was the only one who predicted the orbit sufficiently well to tell them where to look. He apparently had already developed the method when he was aged 18, but he only published it much later. He also, of course, famously defined the normal (Gaussian) distribution, which is closely linked to the least-squares method.

The full story of Gauss, Laplace, Legendre and Ceres.

Maximum likelihood inference

Gaussian noise model:

$$y^t = f(x^t) + \quad _t; \qquad _t \quad \text{Normal}(0; \quad)$$

With conditional density:

$$p\ (y^t j x^t) / \ \exp(\ [y^t \qquad (x^t)]^2 = 2\ ^2)$$

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sklearn

Fitting a model to data

```
from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X, Y)
```

Getting predictions

We can get predictions for all inputs as an array

```
Z = model.predict(X)
```

Statsmodels

Fitting a model to data X, Y

```
import statsmodels.api as sm
Xa = sm.add_constant(X) # adds a constant factor to the data
model = sm.OLS(Y, Xa)
results = model.fit()
```

Getting predictions

The prediction is not just a point!

```
z = results.get_prediction(Xa[t])
z.predicted_mean # This is E[y|x]
```

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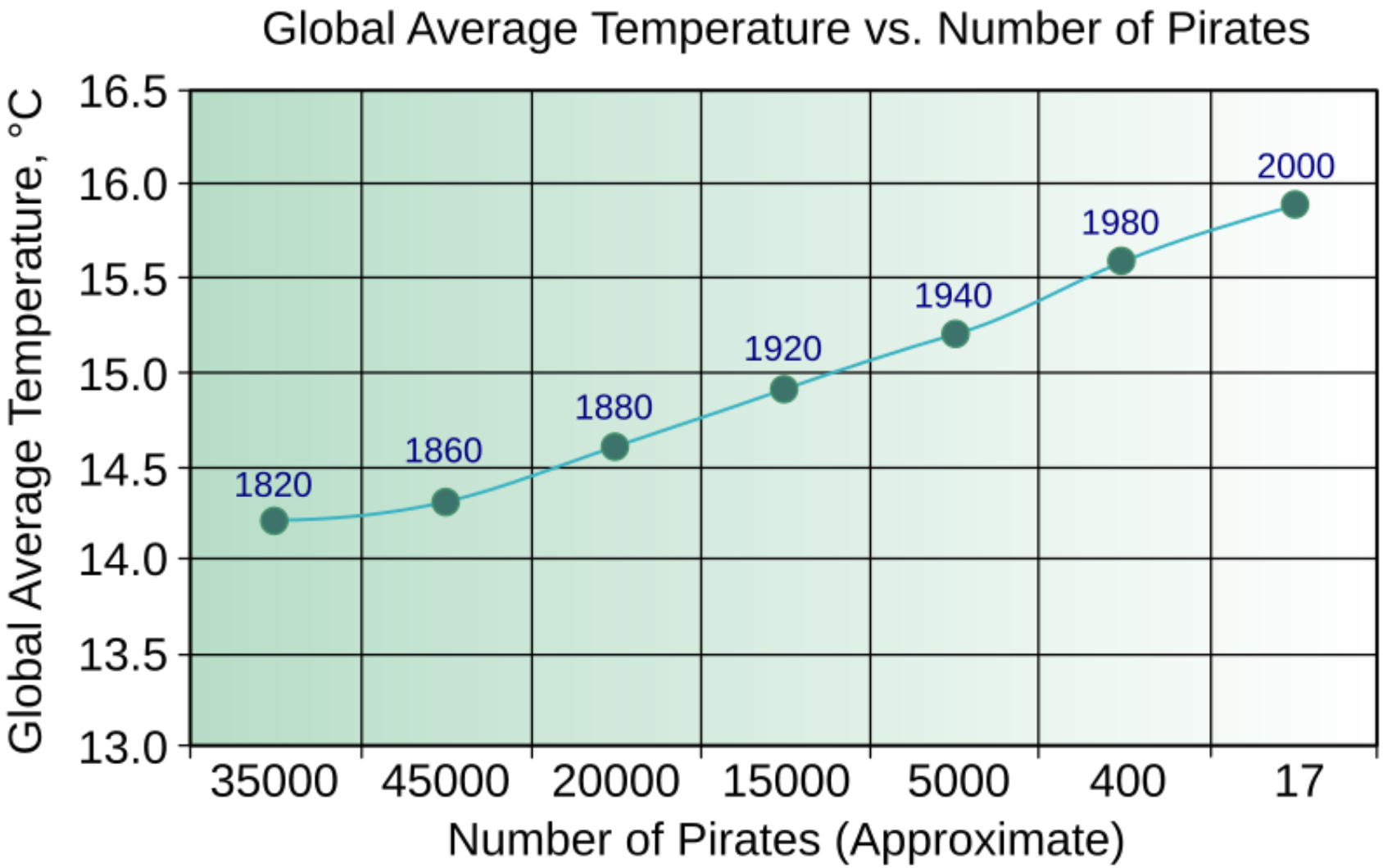
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Pitfalls

- β_i tells us how much y is correlated with $x_{t,i}$
- However, multiple correlations might be evident.
- Some features may be irrelevant
- The relationship may not be linear
- Correlation is not causation

Correlation is not causation



Linear regression exercises

- Exercises 8, 13 from ISLP
- A variant of Ex. 13 but with Y generated independently of X.